CH2 Combinatorial Methods

2.2 Counting principle

Theorem 2.1 (Counting Principle)

If the set E contains n elements and the set F contains m elements,

then, there are n*m ways in which we can choose

Theorem 2.2 (Generalized Counting Principle)

If we have sets $E_1, E_2, E_3, ..., E_n$

then, there are $n_1 \times n_2 \times n_3 \times ... \times n_n$ ways to choice

Example 2.2

In tossing 4 fair dice, P(at least one 3 among these 4 dice) = ?

$N(S) = 6 \times 6 \times 6 \times 6 = 6^4$

$N(A^C) = 5 * 5 * 5 * 5 = 5^4$

Example 2.6 (Standard Birthday Problem)

P(at least two among n people have the same birthdays)=

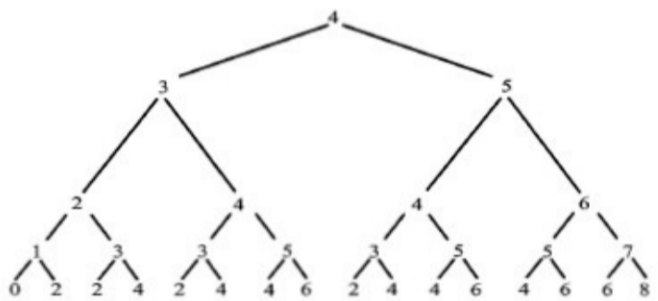
$N(S) = 365 * 365 * ... * 365 = 365^n$

$N(A^C) = 365 * 364 * ... * (365 - n + 1)$

Counting principle: Tree Diagram

Ex 2.9 Mark has $4. He decides to bet $1 on the flip of a fair coin 4 times. What is the probability that (a) he breaks even; (b) he wins money

Ans: (a) 6/16 (b) 5/16



2.3 Permutations

Definition: An **ordered** arrangement of r objects from a set of A containing n objects is called an r-element **permutation** of A, or a **permutation** of the elements of A taken r at a time. The number of r-element permutations of a set containing n objects is denoted by ${}_nP_r$ or $P(n,r)$

Note: ${}_nP_r = P(n,r) = n * (n-1) * (n-2) * \ldots * (n-r+1)$

$= n!/(n-r)!$

${}_nP_n = P(n,n) = n * (n-1) * (n-2) * \ldots * 1 = n!$

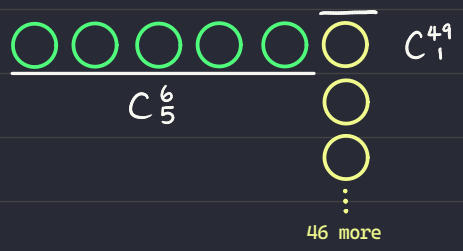
with replacement: 可以重複 (Sample space 不會變小)

2.4 Combinations

Ex 2.19 In the lottery game, players pick 6 integers from 1 to 49, order of selection being irrelevant.

(a) P(Grand prize = 6 matches) = ?

$$P = 1 \div C^{49}_{6}$$



(b) P(2nd prize = 5 matches) = ?

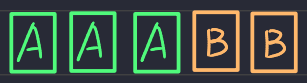
$$P = (C^{6}_{5} \times C^{43}_{1}) \div C^{49}_{6}$$

Ex 2.20 7 cards are drawn from 52 without replacement.

P(at least one of the cards is a king)=?

算 A^c ...

Ex 2.21 5 cards are drawn from 52. P(full house)=?



Theorem

(a) The number of ways n indistinguishable objects can be placed into k distinguishable is

(b) The number of distinct nonnegative integer solutions that the equation $x_1 + x_2 + x_3 + ... + x_k = n$ has is

$$\binom{n+k-1}{n} = \binom{n+k-1}{k-1}$$